

The Best Open Source Software for Network Attached Storage

Network attached storage is a dedicated appliance used for file storage and sharing, which provides advanced functionalities for data storage, access and management. It gives access to files using network file sharing protocols like NFS, SMB or AFP. This article gives a brief on some of the best OSS for network attached storage.



Network attached storage (NAS) was introduced in 1983 with Novell's early file sharing NetWare server operating system and NCP protocol. In the UNIX world, Sun Microsystems' 1984 release of NFS allowed network servers to share their storage space with networked clients. 3Com's 3Server and 3+Share software were the first purpose-built servers (including proprietary hardware, software and multiple disks) for open systems servers, and the company led the segment from 1985 through the early 1990s. 3Com and Microsoft developed the LAN Manager software and protocol to further this new market. Inspired by the success of file servers from Novell, IBM and Sun, several firms developed dedicated file servers. While 3Server was among the first firms to build a dedicated NAS for desktop operating systems, Auspex Systems was one of the first to develop a dedicated NFS server for use in the UNIX market. A group of Auspex engineers split away to create the integrated network appliance 'filer', which supported both Windows and UNIX, in the early 1990s, starting the market for proprietary NAS arrays.

Network attached storage

Network attached storage is regarded as a dedicated server, also termed *appliance*, used for file storage and sharing. As compared to traditional file serving and direct attached storage, NAS provides advanced functionalities in terms of data storage, access and management. NAS systems are networked appliances that contain one or more storage drives, often arranged into logical, redundant storage containers or RAID (redundant array of inexpensive disks). Network attached storage removes the responsibility of file serving

from other servers on the network and provides access to files using network file sharing protocols like NFS (network file system), SMB (server message block) or AFP (Apple file protocol). NAS is regarded as a complete system designed for heavy network systems, which may be processing millions of transactions per minute. It provides a widely supported storage system for any organisation requiring a reliable network system.

Nowadays, organisations search for reliable data storage methods, which can be managed and controlled with their established network systems. NAS allows organisations to store and retrieve humongous amounts of data at an affordable price.

The main NAS components are as follows.

- **NAS protocol:** NAS servers fully support the network file systems and the common interface file systems. NAS also supports numerous different network protocols like SCP and FTP. Reliable communication can be achieved through TCP/IP. The initial purpose of NAS design was file sharing over UNIX across LAN. NAS supports the HTTP/HTTPS protocol, and end users can even download the files using NAS, if connected via a LAN.
- **NAS connections:** Varied mediums are used for establishing connections with NAS servers like Ethernet, optical fibre and even wireless communication standards like 802.11.
- **NAS drives:** SCSI is used as default, and various other mediums like ATA disks, optical disc and magnetic media are also supported via NAS. NAS devices are classified into the following three categories.
- **Computer based NAS:** This is deployed using a computer via FTP/SMB/software server.
- **Embedded system based NAS:** In this category, MIPS